Database creation

```
CREATE DATABASE EcommerceDB;
 GO
 USE EcommerceDB;
 GO
```

Create core tables with constraints

```
 CREATE TABLE ProductPrice (
     PriceID INT PRIMARY KEY IDENTITY(1,1),
     ProductID INT NOT NULL,
     CurrentPrice DECIMAL(10,2) NOT NULL,
     EffectiveDate DATE,
     SysStartTime DATETIME2 GENERATED ALWAYS AS ROW START HIDDEN,
     SysEndTime DATETIME2 GENERATED ALWAYS AS ROW END HIDDEN,
     PERIOD FOR SYSTEM_TIME (SysStartTime, SysEndTime),
     FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
 ) WITH (SYSTEM_VERSIONING = ON);
 GO

SELECT ProductID, CurrentPrice, SysStartTime, SysEndTime
 FROM ProductPrice
 FOR SYSTEM_TIME ALL
WHERE ProductID = 1;
```

	ProductID	CurrentPrice	SysStartTime	SysEndTime
1	1	109.99	2026-09-21 08:23:49.1639895	9999-12-31 23:59:59.9999999
2	1	99.99	2026-09-21 08:23:49.1626793	2026-09-21 08:23:49.1639895

Add JSON columns for metadata

```sql
USE EcommerceDB;
GO
ALTER TABLE Product ADD Metadata JSON;
GO
ALTER TABLE Product ADD MetadataColor AS JSON_VALUE(Metadata, '$.color');
GO
CREATE NONCLUSTERED INDEX IX_Product_Metadata_Color
   ON Product (MetadataColor);
GO
UPDATE Product SET Metadata = N'{"color":"blue","size":"large","material":"cotton"}'
WHERE ProductID = 1;
UPDATE Product SET Metadata = N'{"color":"red","size":"small","material":"silk"}'
WHERE ProductID = 2;
GO

SELECT
   ProductID,
   ProductName,
   JSON_VALUE(Metadata, '$.color') AS Color,
   JSON_VALUE(Metadata, '$.size') AS Size,
   JSON_VALUE(Metadata, '$.material') AS Material
FROM Product
WHERE JSON_VALUE(Metadata, '$.color') = 'blue';
```

Resultados | Mensajes

	ProductID	ProductName	Color	Size	Material
1	1	Wireless Mouse	blue	large	cotton

Create a partitioned order table

```sql
USE EcommerceDB;
GO

SELECT
   $PARTITION.PF_OrderDate(OrderDate) AS PartitionNumber,
   COUNT(*) AS OrdersInPartition,
   MIN(OrderDate) AS MinDate,
   MAX(OrderDate) AS MaxDate
FROM [Order]
GROUP BY $PARTITION.PF_OrderDate(OrderDate);
```

Resultados | Mensajes

	PartitionNumber	OrdersInPartition	MinDate	MaxDate
1	2	2	2025-01-15	2025-02-20
2	3	1	2025-06-10	2025-06-10

Creación de detalles de orden con SEQUENCE

```sql
USE EcommerceDB;
 GO
 CREATE SEQUENCE OrderLineSequence
    START WITH 1
    INCREMENT BY 1;
 CREATE TABLE OrderDetail (
    OrderLineID INT PRIMARY KEY,
    OrderID BIGINT NOT NULL,
    OrderDate DATE NOT NULL,
    ProductID INT NOT NULL,
    Quantity INT NOT NULL,
    UnitPrice DECIMAL(10,2) NOT NULL,
    LineTotal AS (Quantity * UnitPrice),
    CHECK (Quantity > 0),
    CHECK (UnitPrice > 0),
    FOREIGN KEY (OrderID, OrderDate) REFERENCES [Order](OrderID, OrderDate),
    FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
 );
 GO
 INSERT INTO OrderDetail (OrderLineID, OrderID, OrderDate, ProductID, Quantity, UnitPrice)
 VALUES
    (NEXT VALUE FOR OrderLineSequence, 1, '2025-01-15', 1, 2, 99.99),
    (NEXT VALUE FOR OrderLineSequence, 1, '2025-01-15', 2, 1, 149.99),
    (NEXT VALUE FOR OrderLineSequence, 2, '2025-02-20', 1, 3, 99.99);
 GO

 SELECT * FROM OrderDetail;
```

Resultados | Mensajes

	OrderLineID	OrderID	OrderDate	ProductID	Quantity	UnitPrice	LineTotal
1	1	1	2025-01-15	1	2	99.99	199.98
2	2	1	2025-01-15	2	1	149.99	149.99
3	3	2	2025-02-20	1	3	99.99	299.97

Verificar objetos de la base de datos

```sql
USE EcommerceDB;
 GO

 INSERT INTO Product (ProductName, CategoryID, SupplierID, BasePrice, StockQuantity)
 VALUES ('Invalid', 1, 1, -50, 10);

 INSERT INTO OrderDetail (OrderLineID, OrderID, OrderDate, ProductID, Quantity, UnitPrice)
 VALUES
    (NEXT VALUE FOR OrderLineSequence, 1, '2025-01-15', 1, 2, 99.99),
    (NEXT VALUE FOR OrderLineSequence, 1, '2025-01-15', 2, 1, 149.99),
    (NEXT VALUE FOR OrderLineSequence, 2, '2025-02-20', 1, 3, 99.99);
 GO
```

Mensajes

```
Mens. 547, Nivel 16, Estado 0, Línea 6
Instrucción INSERT en conflicto con la restricción CHECK 'CK__Product__BasePri__534D60F1'.
Se terminó la instrucción.

(3 filas afectadas)

Hora de finalización: 2026-09-21T10:42:01.7211011+02:00
```

Verificación BD Completa

```sql
USE EcommerceDB;
GO

SELECT ProductName, JSON_VALUE(Metadata, '$.color') AS Color
FROM Product
WHERE Metadata IS NOT NULL;

SELECT $PARTITION.PF_OrderDate(OrderDate) AS Partition, COUNT(*) AS RecordCount
FROM [Order]
GROUP BY $PARTITION.PF_OrderDate(OrderDate);

SELECT ProductID, CurrentPrice, SysStartTime, SysEndTime
FROM ProductPrice FOR SYSTEM_TIME ALL
ORDER BY ProductID, SysStartTime;
```

Resultados | Mensajes

	ProductName	Color
1	Wireless Mouse	blue
2	Cotton T-Shirt	red

	Partition	RecordCount
1	2	2
2	3	1

	ProductID	CurrentPrice	SysStartTime	SysEndTime
1	1	99.99	2026-09-21 08:23:49.1626793	2026-09-21 08:23:49.1639895
2	1	109.99	2026-09-21 08:23:49.1639895	9999-12-31 23:59:59.9999999
3	2	149.99	2026-09-21 08:23:49.1626793	9999-12-31 23:59:59.9999999